Food & Nutrient Analytics

Using USDA FoodData Central

Processing food data across MongoDB and Hadoop, then comparing the results and performance.

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THE PROJECT

01 Store

02 Analyse

03 Compare

The data is large, and the facts are split across files

A food name, its nutrient amounts and the nutrient definitions are stored separately. We need to connect them before answering useful questions.

01 Organise the records

Link foods to their nutrients without losing the source identity.

02 Calculate useful summaries

Count foods and calculate nutrient totals and averages by food type.

03 Check correctness and performance

Compare the answers across tools and measure search and storage efficiency.

One food can have many nutrient records

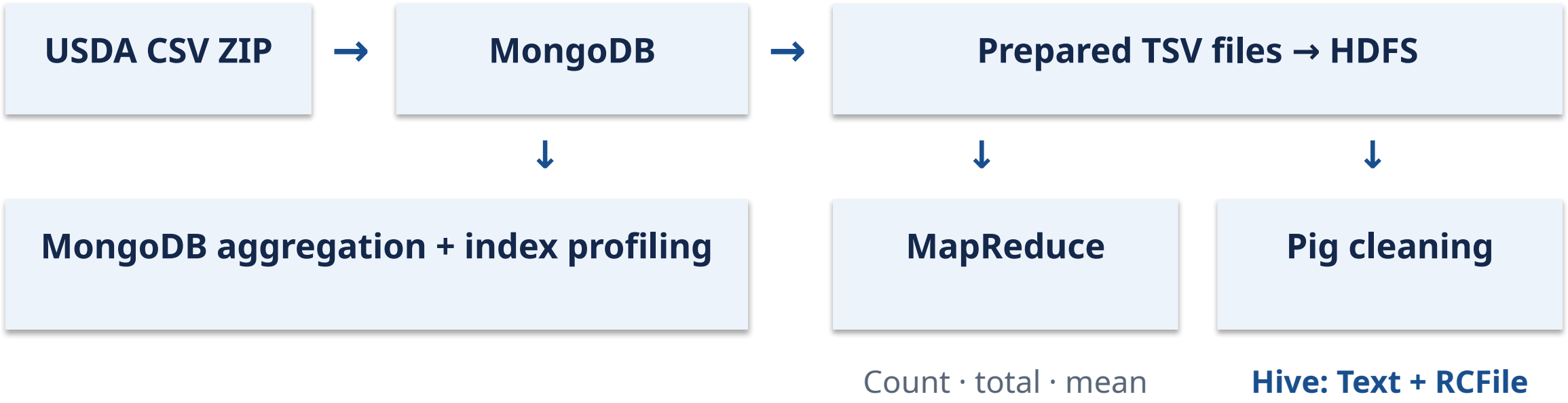
Source file	What it contains	How it connects
food.csv	Food ID, name, food type	fdc_id
food_nutrient.csv	Food ID, nutrient ID, amount	fdc_id + nutrient_id
nutrient.csv	Nutrient name and unit	nutrient_id

Food: WESSON Vegetable Oil → **fdc_id: 1105904** → **Its linked nutrient records**

Source: October 2024 USDA CSV package. Full release: 2.06 million foods.

Presentation test: 50,000 foods + all 1,126,928 linked nutrient records.

The same linked data follows two analysis paths



Check matching answers → Save results and timings → Flask dashboard

MapReduce and Pig read the prepared HDFS data as separate branches.

Each tool produces a result we can demonstrate

Theme	What our program does	What we can show
1 · MongoDB	Import, index and aggregate documents	Raw records → grouped answers
2 · Hadoop	HDFS files; MapReduce on YARN	Files → nutrient summaries
3 · Pig + Hive	Clean data; compare Text and RCFile	Clean rows → query answers

MongoDB pipelines use $match, $group, $project, $unwind and $facet.

HDFS demonstrates mkdir, put, cat, get and rm -r.

Hive runs identical questions on both formats and records time and storage.

The index checks 1 document instead of 50,000

Same MongoDB lookup	Documents examined	Results returned
Collection scan	50,000	1
Indexed lookup	1	1

The answer stays the same; the amount of searching changes.

The 50,000-food test also passed cross-system reconciliation for 1,126,928 linked nutrient records.

Hive comparison: identical answers are checked first, then query time and storage are compared.

These results describe the laptop development run, not a completed physical lab run.

The demo shows the data, the answers and the proof

01 Before analysis

Show a few actual food documents and their linked nutrient records.

02 After analysis

Show food counts, nutrient summaries and Text vs RCFile comparisons.

03 Verify the source

Copy the command beside a result and inspect MongoDB or HDFS in a terminal.

The dashboard also shows analysis progress, logs and the identity of the selected saved run.

One repeatable workflow—from food records to evidence

We connected food and nutrient data, processed it with multiple tools, and checked that the results agree.

01 Data analytics

Food-type counts and nutrient count, total and mean summaries.

02 Performance evidence

MongoDB index profiling and Hive Text/RCFile comparisons.

03 A demonstrable application

Flask dashboard: results, progress and verification commands.

USDA: fdc.nal.usda.gov/download-datasets/ | github.com/Dhanush-singh-g/BDA-Assignment